

Spread Locator

1. What is a Statistical Distribution?

A statistical distribution shows how the values of a dataset are spread and how often different values can occur. It helps us understand the pattern of data.

It gives information about the center of the data, how much the data changes, and whether the data is balanced or not.

2. What is a Q-Q Plot and why is it used?

A Q-Q (Quantile-Quantile) plot is a graph used to check whether the given data follows a particular distribution, mainly the Normal Distribution.

In this graph, the values from our data are compared with the expected values from the theoretical distribution.

Why is it used?

- It is mainly used to check whether the data is normally distributed.
- If points are close to the straight 45-degree line, the data is close to a Normal Distribution.
- If points are far from the line, the data may not be normally distributed.
- A curved pattern can also show that the data is not normal.

Thus, a Q-Q plot gives a simple visual way to check normality.

3. Difference between Discrete and Continuous Distributions

Feature	Discrete Distribution	Continuous Distribution
Definition	Used for values that can be counted separately.	Used for values that can take any value in a range.
Data Values	Usually whole numbers such as 0, 1, 2, 3.	Can have decimal values such as 10.5, 100.75.
Function Used	Probability Mass Function (PMF).	Probability Density Function (PDF).
E-commerce Example	Number of successful transactions in one week.	Amount of money spent in one transaction.

4. What is Bernoulli Distribution?

Bernoulli Distribution is a discrete probability distribution used for one trial where there are only two possible results.

- Success = 1
- Failure = 0

Parameters:

- p = probability of success
- $q = 1 - p$ = probability of failure

PMF Formula: $P(X = k) = p^k(1 - p)^{1-k}$, where $k \in \{0, 1\}$.

E-commerce Example: For one transaction, successful transaction = 1 and failed transaction = 0. Therefore, Bernoulli Distribution can be used.

5. What is Binomial Distribution?

Binomial Distribution is a discrete probability distribution used to find the number of successes in a fixed number of trials.

Conditions:

- The number of trials is fixed.

- Each trial has two possible results.
- The probability of success remains the same.
- The trials are independent.

Parameters:

- n = total number of trials
- p = probability of success in each trial

PMF Formula: $P(X = k) = C(n, k)p^k(1 - p)^{n-k}$, where $k = 0, 1, 2, \dots, n$.

E-commerce Example: If a customer makes 10 orders, we can use Binomial Distribution to find the number of successful orders.

6. Explain Log-Normal Distribution

Log-Normal Distribution is a continuous distribution used for positive values. In this distribution, the logarithm of the values follows a Normal Distribution.

The values of a Log-Normal Distribution are always greater than zero.

Main Characteristics:

- Values are always positive.
- It usually has a long right tail.
- Most values can be small or medium.
- A few values can be very large.

E-commerce Example: Daily transaction amounts may have many small/medium purchases and a few very large purchases, creating a right-skewed pattern.

7. Explain Power Law Distribution

Power Law Distribution shows a relationship where a small number of values can be very large, while many values are small.

Formula: $Y = c \cdot X^{-\alpha}$, where α is the scaling value.

Main Characteristics:

- It can have a long and heavy tail.
- A small number of cases can have very large values.
- Many cases have smaller values.
- It can be used to study an 80/20 type of pattern.

E-commerce Example: A small number of customers may spend very large amounts, while most customers spend less.

8. What is Box-Cox Transform?

Box-Cox transformation is a method used to change data into a form that is closer to a Normal Distribution.

It is mainly useful when the data is skewed or when the variation in the data is not stable.

Formula:

$$y^{\lambda} = (y^{\lambda} - 1)/\lambda, \text{ if } \lambda \neq 0$$

$$y^{\lambda} = \ln(y), \text{ if } \lambda = 0$$

Important Point:

Box-Cox transformation can only be used when all data values are greater than zero: $y > 0$.

9. Explain Poisson Distribution with an Example

Poisson Distribution is a discrete probability distribution used to find the number of times an event happens in a fixed period or area.

It is useful when events happen independently and have a known average rate.

Main Assumptions:

- Events happen independently.
- The average rate remains almost constant.
- Events are counted in a fixed time or area.

PMF Formula: $P(X = k) = (\lambda^k e^{-\lambda}) / k!$

- λ = average number of events
- k = number of events

E-commerce Example: If an online payment system receives an average of 120 transactions per hour, Poisson Distribution can find the probability of receiving a particular number of transactions in one hour.

10. What is Z-score Probability?

A Z-score tells us how far a particular value is from the mean in terms of standard deviation. It tells us whether the value is above or below the mean.

Formula: $Z = (x - \mu) / \sigma$

- x = given value
- μ = mean
- σ = standard deviation

After finding the Z-score, we can use the Standard Normal Distribution table or CDF to find the probability.

Example: To find the probability that a transaction amount is more than ₹5000, first calculate its Z-score and then find the required probability.

11. Difference between PDF and CDF

Feature	PDF	CDF
Definition	Shows how values of a continuous variable are distributed.	Shows total probability up to a given value.
Formula	$f(x) = dF(x)/dx$	$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$
Output	Density value; it can sometimes be greater than 1.	Always between 0 and 1.
Meaning	Shows where data is more concentrated.	Shows probability that X is less than or equal to a given value.

Simple Difference: PDF tells us about the density of values at a particular point, while CDF tells us the total probability up to a particular value.